



PARTS LIST

Semiconductors:

- IC1 - 5V voltage regulator LM7805
- IC2 - PIC16F676 microcontroller
- T1-T3 - BC547 NPN transistor
- D1-D5 - 1N4007 rectifier diode
- IR1 - TSOP 1738 receiver module

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R3 - 10-kilo-ohm

Capacitors:

- C1 - 2200 μ F, 25V electrolytic
- C2, C3 - 10 μ F, 25V electrolytic
- C4, C5 - 22pF ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 12V-0-12V, 1A secondary transformer
- CON1 - 2-pin connector
- L1-L3 - 230V AC lamp (any wattage)
- RL1-RL3 - 5V, single changeover relay
- NEC remote

Program stores all the bytes in the array and then decodes the third byte to use for comparison.

The simple programming statement 'switch' case is used to detect and control the home appliances.

Do not forget to set fuse bits be-

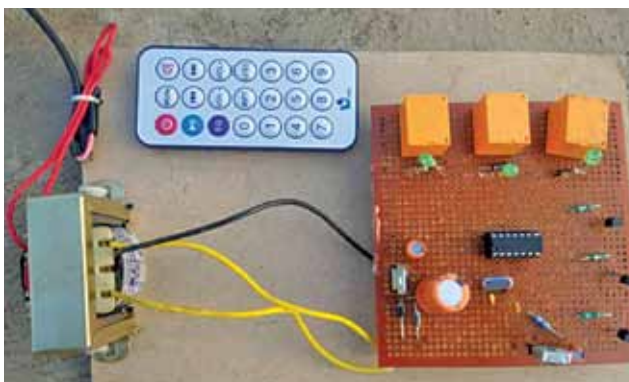


Fig. 11: Author's prototype

fore programming PIC16F676 using PIC K150 programmer. The program will not work without setting the fuse bits. The fuse bit settings are shown in Fig. 8.

Construction and testing

An actual-size PCB layout for the automation system is shown in Fig. 9 and its component layout in Fig. 10. After assembling the circuit on

PCB, enclose it in a suitable box. Receiver IR1 should be fixed on the front panel, so that the remote can focus on it to activate any load connected to the respective relay. CON1 should be fixed on back side of the cabinet. All the three 230V

AC appliances are connected to the circuit using two wires and placed at their location. The author's prototype wired on a general-purpose PCB is shown in Fig. 11. **EFY**



Pamarthi Kanakaraja is an assistant professor (R&D cell) at K.L. University, Vaddeswaram, Guntur district, Andhra Pradesh

Job Guaranteed.

You don't need a training institute. Just do it yourself.

WORK WITH OPEN SOURCE.

www.opensourceforu.com

Demand for Open Source is sky rocketing. Be it for managing IT infrastructure or development of software—Open Source experts are what organisations are seeking. Develop expertise in an Open Source stack, and watch your career graph zoom.

You don't need a training institute for this. You can do it yourself. And, *Open Source For You* can be your friend and a guide through this journey.

TO READ OUR PRINT EDITION Visit: <https://subscribe.efyindia.com>

TO READ OUR EZINE EDITION Visit: <https://ezine.ifymag.com>

WORLD'S LEADING PUBLICATION ON OPEN SOURCE

Looking for marketing solutions to engage with cutting edge techies? Contact us at growmybiz@efy.in OR call us at +91-9811155335.

